

Ecuaciones Principales.

$$V_e(t) = R i_0(t) + L \frac{d[i_2(t) - i_z(t)]}{dt} + R [i_1(t) - i_z(t)]$$

$$V_s(t) = R i_z(t) + \frac{1}{C} \int i_z(t) dt$$

• Transformada de Laplace

$$V_e(s) = R I_1(s) + L s [I_1(s) - I_z(s)] + R [I_1(s) - I_z(s)]$$

$$V_s(s) = R I_z(s) + \frac{I_z(s)}{Cs}$$

• Proc. Algebraico

$$V_e(s) = (Ls + 2R) I_1 - (Ls + R) I_z(s)$$

$$V_s(s) = \left(R + \frac{1}{Cs}\right) I_z(s) = \frac{CRs + 1}{Cs} I_z(s)$$

$$\frac{3CLR^2s^2 + (5CR + 1)s + 2R}{CLs^2 + CRs}$$

• Func. transferencia:

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{CR+1}{Cs}}{\frac{3CLR^2s^2 + (5CR+1)s + 2R}{CLs^2 + CRs}}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{CLR^2s^2 + (CR^2 + L)s + R}{3CLR^2s^2 + (5CR + 2L)s + 2R}$$

Error de estado est.

$$e(s) = \lim_{s \rightarrow s_0} \left[1 - \frac{CLR(\cos^2 + (LR+L)\cos + R)}{5CLR\cos^2 + (5CR+L)\cos + 2R} \right]$$

$$e(s) = \lim_{s \rightarrow s_0} 1 - 0.05 = 0.5$$

$$e(s) = 0.5V$$

• Estabilidad L. Abierto.

$$\begin{aligned} a &= 3CLR \\ b &= 5CR^2 + L \\ c &= 2R \end{aligned}$$

$$\begin{aligned} R &= 2.2K \\ C &= 470 \mu F \\ L &= 68 mH \end{aligned}$$

$$s_{1,2} = \frac{-(5CR^2 + L) \pm \sqrt{(5CR^2 + L)^2 - 4(3CLR)(2R)}}{2(3CLR)}$$

$$s_1 = 0.3868476579$$

$$s_2 = -53921.50415$$

Sobreamortiguado

• Modelo de EC. integrodiferenciales

$$V_e(t) = 2R i_1(t) + L \frac{d}{dt} [i_1(t) - i_2(t)] - R i_2(t)$$

$$i_2(t) \left[\frac{d}{dt} [i_1(t) - i_2(t)] \right] + R i_2(t) = -R i_2(t) + R i_1(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$I_1(t) = \int L dE$$